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## 1. SQL Databases (Relational Databases)

**Examples:** MySQL, PostgreSQL, Oracle, MS SQL Server

**Best For:** Structured data, strong consistency, ACID transactions

♦ **Use Cases:**

- ✓ Banking & finance (e.g., transactions, accounts)
- ✓ E-commerce (orders, inventory)
- ✓ Enterprise applications (ERP, CRM)

♦ **Advantages:**

- ✓ Strong ACID compliance (Atomicity, Consistency, Isolation, Durability)
- ✓ Complex queries with SQL (joins, aggregations)
- ✓ Well-defined schemas ensure data integrity

♦ **Disadvantages:**

- ✗ Vertical scaling (harder to scale horizontally)
  - ✗ Fixed schema (less flexibility for dynamic data models)
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## 2. NoSQL Databases (Non-Relational)

**Examples:** MongoDB, Cassandra, Redis, DynamoDB

**Best For:** Unstructured/semi-structured data, high scalability, real-time applications

♦ **Types of NoSQL Databases:**

- **Document-based:** JSON-like storage (MongoDB)
- **Key-Value stores:** Fast lookups (Redis, DynamoDB)
- **Column-family stores:** Big data & analytics (Cassandra, HBase)
- **Graph databases:** Relationship-heavy data (Neo4j)

- ♦ **Use Cases:**

- ✓ Social networks (relationships, recommendations)
- ✓ IoT & real-time analytics
- ✓ Big Data applications

- ♦ **Advantages:**

- ✓ Schema flexibility (easy to modify structure)
- ✓ High availability & partition tolerance (CAP theorem)
- ✓ Scales horizontally (distributed systems)

- ♦ **Disadvantages:**

- ✗ Weaker consistency (most NoSQL systems use eventual consistency)
  - ✗ Lack of complex joins (requires denormalization)
  - ✗ Querying & indexing can be less efficient compared to SQL
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## 3. Other Databases

### NewSQL Databases

**Examples:** Google Spanner, CockroachDB, TiDB

- ♦ **Best for:** Scalability of NoSQL + ACID compliance of SQL
- ♦ **Use Case:** Large-scale transactional systems requiring high availability

### Time-Series Databases

**Examples:** InfluxDB, TimescaleDB

- ♦ **Best for:** IoT, logs, and time-stamped data (metrics, analytics)
- ♦ **Use Case:** Monitoring systems, financial market data

### Graph Databases

**Examples:** Neo4j, ArangoDB

- ♦ **Best for:** Relationship-intensive data
- ♦ **Use Case:** Fraud detection, recommendation engines

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## How to Choose?

| Criteria         | SQL (Relational)          | NoSQL (Non-Relational)         | NewSQL/Other               |
|------------------|---------------------------|--------------------------------|----------------------------|
| Data Structure   | Structured (Fixed Schema) | Semi-structured / Unstructured | Structured/Semi-structured |
| Scalability      | Vertical Scaling          | Horizontal Scaling             | Both                       |
| Consistency      | Strong (ACID)             | Eventual Consistency (Mostly)  | Strong (ACID)              |
| Query Complexity | SQL (Joins, Aggregations) | Limited Joins (Denormalized)   | SQL-like queries           |
| Use Case         | Banking, ERP, E-commerce  | Real-time, IoT, Big Data       | Distributed Transactions   |